

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks.
(BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

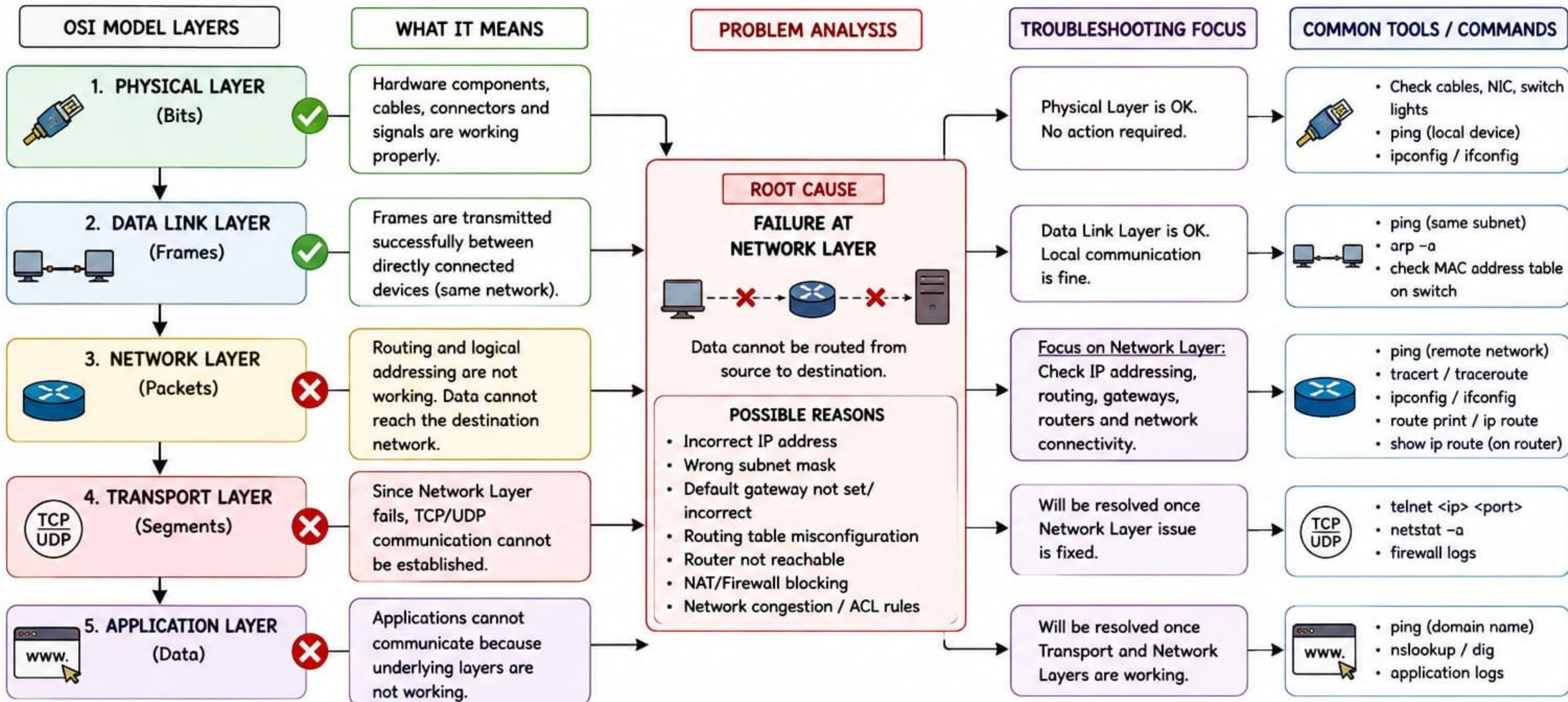
6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

NETWORK ISSUE – CONCEPT MAP

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Status: Physical  → Data Link  → Network  → Transport  → Application 

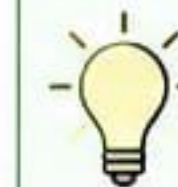


HOW THE CONCEPT MAP HELPS

-  Provides a layer-by-layer view of the network communication.
-  Helps in quickly identifying the layer where the issue starts.
-  Avoids checking all layers blindly and saves time.
-  Focuses troubleshooting on the exact cause (Network Layer).
-  Ensures efficient isolation and faster resolution.

TROUBLESHOOTING APPROACH (STEP-BY-STEP)

- 1 Start from Physical Layer → verify connectivity.
- 2 Move to Data Link Layer → ensure local communication.
- 3 Reach Network Layer → issue found here.
- 4 Check IP, routing, gateways, routers, ACLs, firewalls.
- 5 Fix the issue → higher layers will start working automatically.



SUMMARY

Since the failure starts at the Network Layer, all upper layers (Transport & Application) also fail. Fix the Network Layer issue to restore end-to-end communication.

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships 20	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues 20	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships 10	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis 15	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map 15	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

70 **80**
100